Section 18 Report Drafting Guide
Annual Climate and Year-Round Operating Layer

April 21, 2026

1 Purpose

This document is a report-writing guide for Section 18 of `docs/execution_docs/physics/physics_and_math_ref` ("Annual climate and year-round operating layer", Eqs. 229–251). It maps each variable from the equations to:

- its JSON source in `Heat Transfer Study/study_config.json`,
- its runtime variable in `Heat Transfer Study/heat_transfer_study.py`,
- and the exported outputs in `Heat Transfer Study/outputs/year_round/study_summary.txt`.

2 Identity Clarifications For Implicit Symbols

To remove ambiguity in the Section 18 equations:

- m: month index (1 to 12).
- d: day-of-year evaluation index (1 to P).
- P: annual day period ($P = 365$ in the present implementation).
- d_i: day-of-year reference day for reference point i (month midpoint in current implementation).
- d_m: month-midpoint day-of-year for month m. With one reference point per month, $d_m \equiv d_i$.
- L_m: number of days in month m from `daysPerMonth`.
- i in Eqs. (229)–(230): index over monthly reference points.
- r in Eqs. (232), (236): day-of-month index inside one month for event placement. This is a different role from kernel index i.
- y_i: selected reference-value family at reference point i. It is one of:
  - $y_i = \bar{T}_i$ for the daily baseline ambient regression pass,
  - $y_i = \sigma_i$ for the daily ambient-spread regression pass.
- $\hat{y}(d)$: kernel-regressed daily estimate from the active y_i pass.
- $w_i(d)$: Gaussian kernel weight for reference point i at day d.

- h: Gaussian kernel bandwidth in days.
- $\mathrm{LR}(\cdot)$: largest-remainder operator used to convert expected monthly event counts to integer allocations.
- $s_{f,m}$, $r_{f,m}$, $\mathcal{S}_{f,m}$: freeze placement spread, repulsion length scale, and selected-day set for month m.
- $s_{h,m}$, $r_{h,m}$, $\mathcal{S}_{h,m}$: heat-wave start placement spread, repulsion length scale, and selected-start set for month m.
- q: heat-wave spell index within month m, ordered by start day.
- $\Delta T_{h,m}^{\mathrm{low}}, \Delta T_{h,m}^{\mathrm{high}}$: NOAA-derived lower/upper bounds for heat-wave spell anomaly above $\bar{T}_m$.
- $\sigma_{\Delta T,h,m}$: configured or fallback spread for bounded spell-anomaly sampling in month m.
- $\Delta T_{h,m,q}$: sampled anomaly assigned to spell q, bounded to $[\Delta T_{h,m}^{\mathrm{low}}, \Delta T_{h,m}^{\mathrm{high}}]$.
- f_{day}: generic clipped-duty function in Eq. (244).
- $\dot{Q}_{\mathrm{load}}$, $\dot{Q}_{\mathrm{nom}}$: generic load/capacity placeholders inside Eq. (244); these map to heating or cooling quantities depending on active mode.
- $Q_{\mathrm{to\,bed},h}(T_\infty(d))$: signed net heat-to-bed from the heating re-solve at day ambient.
- $Q_{\mathrm{to\,bed},c}(T_\infty(d))$: signed net heat-to-bed from the cooling re-solve at day ambient.
- $Q_{h,\mathrm{cap}}(d)$, $Q_{c,\mathrm{cap}}(d)$: positive heating/cooling capacities used in duty equations, where $Q_{h,\mathrm{cap}}(d) = Q_{\mathrm{to\,bed},h}(T_\infty(d))$ and $Q_{c,\mathrm{cap}}(d) = -Q_{\mathrm{to\,bed},c}(T_\infty(d))$.
- $P_h(T_\infty(d))$: heating electrical power from the day-ambient heating re-solve.
- $P_{\mathrm{spot\,cooler}}(d)$, $P_{\mathrm{assist\,blower}}(d)$: cooling electric-power components from the day-ambient cooling re-solve.
- $P_{h,\mathrm{elec}}(d)$, $P_{c,\mathrm{elec}}(d)$: mode electric powers used in day-energy equations, with $P_{h,\mathrm{elec}}(d) = P_h(T_\infty(d))$ and $P_{c,\mathrm{elec}}(d) = P_{\mathrm{spot\,cooler}}(d) + P_{\mathrm{assist\,blower}}(d)$.
- $D(d)$ in Eq. (251): active duty fraction for the day, equal to $D_h(d)$ on heating days or $D_c(d)$ on cooling days.

3 Temperature Variables: Explicit Definitions and Sources

- $\bar{T}_m$: monthly Connecticut mean ambient from `climate_data.monthlyMean_C`.
- σ_m: monthly ambient spread from `climate_data.monthlyStd_C`.
- $T_{\infty,\mathrm{base}}(d)$: periodic Gaussian-kernel regression of $\bar{T}_m$ over day-of-year.
- $\sigma(d)$: periodic Gaussian-kernel regression of σ_m over day-of-year.
- $T_{\infty,\mathrm{preH}}(d)$: intermediate profile after freeze-day overrides (input to 18.3 heat-wave overlay).

- $T_{\infty,\mathrm{heatRaw}}(d)$: optional diagnostic track exported by code; the governing 18.3 equation below is written directly in combined form for $T_\infty(d)$.
- $T_m(d)$: month-local ambient profile during event-overlay construction.
- $T_\infty(d)$: final day ambient used in load/capacity re-solves.
- $T_{h,\mathrm{set}}, T_{c,\mathrm{set}}$: resolved heating/cooling setpoints from annual config (or inherited design defaults).
- $T_b(d), T_t(d)$: bottom/top bed-node temperatures from Eq. (247), solved using $Q_{\mathrm{bed}}(d)$ and $T_\infty(d)$.
- Heat-wave spell threshold temperature: $\bar{T}_m + \Delta T_{h,m,q}$ for spell q in month m.
- NOAA derivation artifacts for spell-anomaly bounds:
  - `Heat Transfer Study/data/climate/noaa_daily_summaries_USW00014740_2016_2025.json`,
  - `Heat Transfer Study/data/climate/noaa_heatwave_spell_anomaly_bounds_USW00014740_2016_2025.`
  - `Heat Transfer Study/tools/derive_noaa_heatwave_spell_bounds.py`.

4 Repository Scope Relevant To Section 18

Main execution code

- `Heat Transfer Study/heat_transfer_study.py`
- Section 18 core functions are concentrated at lines 5894–6740 (current repository state).

Main configuration inputs

- `Heat Transfer Study/study_config.json`
- Key blocks: `year_round_analysis`, `year_round_analysis.climate_data`, `cost_analysis`.

Primary generated outputs

- `Heat Transfer Study/outputs/year_round/study_summary.txt`
- `Heat Transfer Study/outputs/year_round/year_round_energy_cost_analysis.png`

Reference documentation

- `docs/execution_docs/physics/physics_and_math_reference.tex`
- Section 18 starts at line 1876 and includes Eqs. 229–251.

5 Execution Chain (What Happens In Code)

1. **Load and normalize annual config**: `year_round_analysis_config(...)` (L5894–).
2. **Build representative daily climate**: `build_representative_climate_year(...)` (L6158–), using:
   - `largest_remainder_allocation(...)` (L6009–),
   - `sample_truncated_normal(...)` (spell-level anomaly sampling),
   - `gaussian_weighted_day_selection(...)` (L6042–),
   - `heatwave_day_mask(...)` (L6082–),
   - `circular_gaussian_kernel_regression(...)` (L6141–).
3. **Run daily thermal and duty solve**: `simulate_year_round_day_task(...)` (L6389–).
4. **Aggregate annual and monthly outputs**: `build_year_round_payload(...)` (L6498–).
5. **Write report output text**: `write_year_round_summary(...)` (L6687–).

6 Variable Connection Map For Eqs. 229–251

Math Symbol	Equation Role	JSON Source	Code Variable / Expression	Output Field
$\bar{T}_m$	monthly mean ambient reference value	`climate_data.monthlyMeanC`	`monthly_mean`, `mean_c` (L6120, L6192)	`monthlyMean_C`
σ_m	monthly ambient spread reference value	`climate_data.monthlyStdC`	`monthly_std`, `std_c` (L6121, L6193)	`monthlyStd_C`, `ambientSigma_C`
h	Gaussian bandwidth (days)	`climate_data.gaussianKernelBandwidthDays`	`bandwidth_days` (L6169)	`gaussianBandwidth_days`
$\hat{y}(d)$	smooth daily baseline from reference values (Eq. 229)	same as above	`daily_mean_regressed` (L6171)	`ambientBase_C`
$\delta_i(d)$	circular day distance (Eq. 230)	period implied by days/month	`deltas = min(d, P-d)` (L6109–L6110)	not directly written
F_m	expected freeze days/month	`freeze.averageDaysPerMonth`	`freeze_expected` (L6124)	`freezeDaysExpected`
$n_{f,m}$	allocated integer freeze days (Eq. 231)	derived from F_m	`freeze_alloc` (L6157)	`freezeDaysAllocated`, annual freeze total
$\mu_{f,m}$	mean freeze day-of-month	`freeze.meanDayOfMonth`	`freeze_mean_day` (L6126)	`freezeMeanDayOfMonth`
$\Delta T_{f,m}$	freeze anomaly below monthly mean	`freeze.meanAnomalyBelowMonthlyMeanC`	`freeze_anomaly` (L6125)	`freezeAnomalyBelowMean_C`

Math Symbol	Equation Role	JSON Source	Code Variable / Expression	Output Field
$T_\infty(d)$ (freeze-adjusted)	freeze override (Eq. 234)	derived from baseline + anomaly	`ambient_profile[freeze_mask] = min(...)` (L6207–L6211)	`ambient_C`, `freezeDay`
S_m	expected heat-wave spell starts/month	`heat_wave.averageSpellStartsPerMonth`	`heat_expected_starts` (L6127)	`heatWaveSpellStartsExpec`
H_m	expected heat-wave days/month	`heat_wave.averageDaysPerMonth`	`heat_expected_days` (L6128)	`heatWaveDaysExpected`
$n_{s,m}, n_{h,m}$	allocated starts and days (Eq. 235)	derived from S_m, H_m	`heat_start_alloc`, `heat_day_alloc` (L6158–L6159)	`heatWaveSpellStartsAlloc` `heatWaveDaysAllocated`
$\mu_{h,m}$	mean heat-wave start day	`heat_wave.meanStartDayOfMonth`	`heat_mean_start` (L6130)	`heatWaveMeanStartDayOfMo`
$\Delta T_{h,m}$	monthly mean spell anomaly above monthly mean	`heat_wave.meanAnomalyAboveMonthlyMean` (NOAA-derived spell means)	`heat_anomaly_mean_C`	`heatWaveAnomalyAboveMean`
$\Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}$	monthly spell-anomaly bounds	`heat_wave.minSpellAnomalyAboveMonthlyMean`, `maxSpellAnomalyAboveMonthlyMean` (loaded from derived NOAA bounds JSON)	`heat_anomaly_low_C`, `heat_anomaly_high_C`	`heatWaveAnomalyMin_C`, `heatWaveAnomalyMax_C`
$\sigma_{\Delta T,h,m}$	spell-anomaly sampling spread	`heat_wave.spellAnomalyStdDev` (NOAA-derived spell spread; fallback from bounds if absent)	`heat_std`	`heatWaveAnomalyStd_C`
$\Delta T_{h,m,q}$	sampled anomaly for spell q (bounded normal draw)	derived from monthly mean/bounds/std + seed	`sample_truncated_normal(...)`	`heatWaveSpellAnomalies_C` day field `heatWaveSpellAnomalyAbov`
$T_\infty(d)$ (heat-adjusted)	heat-wave override (Eq. 238) with spell-specific anomaly	derived from baseline + $\Delta T_{h,m,q(d)}$	spell-loop max clamp on each selected spell window	`ambient_C`, `heatWaveDay`, `heatWaveSpellIndex`
$T_{h,\text{set}}$, $T_{c,\text{set}}$	heating/cooling setpoints	`year_round_analysis.heatingSetpoint_C`, `coolingSetpoint_C` (or resolved defaults)	`heating_setpoint_C`, `cooling_setpoint_C` (L6388–L6389)	printed as resolved setpoints in summary
$Q_{\text{req},h}(d)$	required heating load (Eq. 239)	derived from bin model and ambient	`required_heat_W` (L6287)	`requiredHeat_W`
$Q_{\text{req},c}(d)$	required cooling load (Eq. 240)	derived from bin model and ambient	`required_cooling_W` (L6288)	`requiredCooling_W`
$Q_{h,\text{cap}}(d)$	heating capacity at day ambient (Eq. 241)	reference point + ambient re-solve	`nominal_capacity_W` from heating state (L6298–L6300)	`nominalCapacity_W`
$P_{h,\text{elec}}(d)$	heating electric power (Eq. 241)	same	`nominal_power_W` from `totalPower_W` (L6300)	`nominalPower_W`
$Q_{c,\text{cap}}(d)$	cooling capacity at day ambient (Eq. 242)	cooling point + ambient re-solve	`nominal_capacity_W = max(-QtoBed, 0)` (L6312)	`nominalCapacity_W`

Math Symbol	Equation Role	JSON Source	Code Variable / Expression	Output Field
$P_{c,\mathrm{elec}}(d)$	cooling electric power (Eq. 242)	cooling solve outputs	`spotCoolerPower_W + assistBlowerPower_W` (L6313–L6314)	`nominalPower_W`
$D_h^*(d)$, $D_c^*(d)$	required duty fraction (Eq. 243)	derived ratio load/capacity	`duty_required` (L6320)	`dutyRequired`
$D_h(d)$, $D_c(d)$	clipped applied duty (Eqs. 244–245)	bounded physically to $[0,1]$	`duty_applied` (L6321)	`dutyApplied`
$Q_{\mathrm{bed}}(d)$	applied bed load (Eq. 246)	mode + duty + signed capacity	`q_bed_applied_W` (L6324–L6336)	`appliedQtoBed_W`
$T_b(d), T_t(d)$	bottom and top node temperatures (Eq. 247)	two-node model + applied bed load	`tb_C, tt_C` (L6326, L6331, L6336)	`bottom_C`, `top_C`
$E_h(d), E_c(d)$	daily mode-specific kWh (Eq. 248)	power, duty, 24 h conversion	`heating_kWh`, `cooling_kWh` (L6327, L6333)	`heating_kWh`, `cooling_kWh`
E_{day}	total daily kWh (Eq. 249)	sum of mode energies	`total_kWh` (L6340)	`total_kWh`
c_{elec}	electricity price (Eq. 250)	`cost_analysis.electricity_price_usd_per_kWh`	`electricity_price_usd_per_kWh` (L6390)	printed in annual summary
$C(d)$	daily cost (Eq. 250)	$c_{\mathrm{elec}}E_{\mathrm{day}}$	`cost_usd` (L6341)	`cost_usd`, cumulative cost
$m_{\mathrm{water}}(d)$	duty-scaled daily water replacement (Eq. 251)	nominal water loss rate from day mode	`water_kg = water_rate_kg_day * duty_applied` (L6342)	`water_kg`

7 Sequential Variable Feed (Section 18 Format)

This section mirrors the Section 18 structure, but emphasizes the feed-forward dependency of variables.

18.1 Climate reference values and regression baseline (Eqs. 229–230)

Inputs:

- month lengths: L_m from `daysPerMonth`,
- month-midpoint reference days: d_i (equivalently d_m),
- monthly means: $\bar{T}_m$,
- monthly standard deviations: σ_m,
- kernel bandwidth: h.

Sequential feed:

$$\{\bar{T}_m,\ d_i,\ h\} \to \delta_i(d),\ w_i(d) \to \hat{T}_{\infty,\mathrm{base}}(d)$$

$$\{\sigma_m,\ d_i,\ h\} \to \delta_i(d),\ w_i(d) \to \hat{\sigma}(d)$$

In explicit equation form:

$$T_{\infty,\text{base}}(d) = \frac{\sum_{i=1}^{12} w_i(d)\,\bar{T}_i}{\sum_{i=1}^{12} w_i(d)}, \qquad w_i(d) = \exp\left[-\frac{1}{2}\left(\frac{\delta_i(d)}{h}\right)^2\right],$$

$$\delta_i(d) = \min(|d - d_i|,\ 365 - |d - d_i|)\,,$$

where d_i is month-i midpoint day-of-year and $\bar{T}_i$ is NOAA monthly mean for month i.

Outputs passed forward:

- baseline daily ambient: $T_{\infty,\text{base}}(d)$,
- daily sigma context: $\sigma(d)$.

18.2 Freeze-day allocation and anomaly override (Eqs. 231–234)

Inputs:

- expected freeze days F_m,
- mean freeze day $\mu_{f,m}$,
- freeze anomaly center $M_{f,m} \equiv \Delta T_{f,m}$,
- freeze anomaly bounds $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$,
- freeze anomaly spread $\sigma_{\Delta T,f,m}$,
- freeze anomaly model key κ_f,
- freeze anomaly RNG seed s_f,
- $T_{\infty,\text{base}}(d)$ from 18.1.

Sequential feed:

$$F_m \to n_{f,m} = \text{LR}(F_m)$$

$$\{n_{f,m}, \mu_{f,m}\} \to w_{f,m}(i),\ \tilde{w}_{f,m}(i) \to \mathcal{S}_{f,m}$$

Here $s_{f,m}$ is the month-m freeze-day spread parameter and $r_{f,m}$ is the repulsion length scale.

$$s_{f,m} = \max\left(2, \frac{L_m}{6}\right), \qquad r_{f,m} = \max\left(1.5, \frac{L_m}{14}\right),$$

where L_m is the number of days in month m.

$$a_{f,m} = \Delta T^{\text{low}}_{f,m}, \quad b_{f,m} = \Delta T^{\text{high}}_{f,m}, \quad \mu^{\star}_{f,m} = \text{clip}(M_{f,m}, a_{f,m}, b_{f,m}).$$

Here $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$ are anomaly-magnitude bounds (not temperatures), from

`climate_data.freeze.minAnomalyBelowMonthlyMean_C`$[m]$, `climate_data.freeze.maxAnomalyBelowMont`

Historical NOAA derivation in month m: for each freeze day j,

$$a_{m,j} = \max\left(0, \bar{T}_m - T^{\text{freeze day}}_{m,j}\right),$$

$$\Delta T^{\text{low}}_{f,m} = \min_j a_{m,j}, \qquad \Delta T^{\text{high}}_{f,m} = \max_j a_{m,j}.$$

So $\mu^\star_{f,m}$ is the bounded (clipped) freeze-anomaly center for month m.

$$c^{\text{raw}}_{f,m} = 3\mu^\star_{f,m} - a_{f,m} - b_{f,m}, \qquad c_{f,m} = \text{clip}(c^{\text{raw}}_{f,m}, a_{f,m}, b_{f,m}).$$

Here $c^{\text{raw}}_{f,m}$ is the unconstrained triangular-mode target: it enforces

$$\frac{a_{f,m} + b_{f,m} + c^{\text{raw}}_{f,m}}{3} = \mu^\star_{f,m}$$

before clipping to the physical interval $[a_{f,m}, b_{f,m}]$. Why: this mean-matching keeps stochastic freeze anomalies centered on the intended monthly value instead of introducing systematic bias.

$$t_{f,m}(x) = \begin{cases} \dfrac{2(x - a_{f,m})}{(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & a_{f,m} \le x \le c_{f,m}, \\ \dfrac{2(b_{f,m} - x)}{(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & c_{f,m} \le x \le b_{f,m}, \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{norm}(s) = \text{lower}\Big(\text{replace}\left(\text{replace}(s, \texttt{"-"}, \texttt{"_"}), \texttt{" "}, \texttt{"_"}\right)\Big), \quad \kappa_f = \text{norm}(\texttt{freeze.anomalySamplingModel}).$$

$$\texttt{isDeterministic}_f = \begin{cases} 1, & \kappa_f = \texttt{off} \text{ or } \texttt{none} \text{ or } \texttt{disabled} \text{ or } \texttt{deterministic} \text{ or } \texttt{deterministic_mean}, \\ 0, & \text{otherwise.} \end{cases}$$

$$\texttt{isTriangular}_f = \begin{cases} 1, & \kappa_f = \texttt{triangular} \text{ or } \texttt{triangular_mean_matched} \text{ or } \texttt{triangular_between_monthly_min_} \\ 0, & \text{otherwise.} \end{cases}$$

Conceptually:

- κ_f is the freeze-anomaly mode label after text normalization. It acts as a branch selector.
- $\texttt{isDeterministic}_f = 1$: deterministic month-level assignment, so every selected freeze day gets $\mu^\star_{f,m}$.
- $\texttt{isTriangular}_f = 1$: stochastic triangular assignment, so each selected freeze day draws its own $X_{f,m,q} \in [a_{f,m}, b_{f,m}]$.
- $\text{norm}(\cdot)$ exists only so different spellings of the same mode map to the same branch.
- These are yes/no control flags, not random-variable samples.

$$\Delta T_{f,m,q} = \begin{cases} \mu^\star_{f,m}, & \texttt{isDeterministic}_f = 1, \\ X_{f,m,q}, \quad X_{f,m,q} = \text{Triangular}(a_{f,m}, c_{f,m}, b_{f,m}), & \texttt{isTriangular}_f = 1 \text{ and } b_{f,m} > a_{f,m}, \\ \mu^\star_{f,m}, & \text{otherwise.} \end{cases}$$

$X_{f,m,q} = \text{Triangular}(a_{f,m}, c_{f,m}, b_{f,m})$ for the default freeze model, then clip to $[a_{f,m}, b_{f,m}]$.

$$\Delta T_{f,m,q} \leftarrow \text{clip}(\Delta T_{f,m,q}, a_{f,m}, b_{f,m}).$$

Sort selected freeze days as $d_{f,m,1} < \cdots < d_{f,m,n_{f,m}}$,

$$A_{f,m}(r) = \begin{cases} 0, & r \notin \mathcal{S}_{f,m}, \\ \Delta T_{f,m,q}, & r = d_{f,m,q} \text{ for some } q \in \{1, \ldots, n_{f,m}\}, \end{cases} \qquad r = 1, \ldots, L_m.$$

Important: $A_{f,m}(r)$ is a deterministic allocation map, not a random draw function.

Randomness, when enabled, is only in generating $\Delta T_{f,m,q}$ ($\texttt{isTriangular}_f = 1$), then q is assigned to ordered d_f

$$r_m(d) = \text{day-of-month index for day-of-year } d \text{ in month } m.$$

$$T^{\text{th}}_{f,m}(d) = \bar{T}_m - A_{f,m}(r_m(d)).$$

Meaning: $T^{\text{th}}_{f,m}(d)$ is the freeze-threshold ambient on day d (the clamp value used by freeze override).

$$\{T_{\infty,\text{base}}(d), \bar{T}_m, \Delta T_{f,m,q}, \mathcal{S}_{f,m}\} \to T_{\infty,\text{preH}}(d) = \min\Big(T_{\infty,\text{base}}(d), T^{\text{th}}_{f,m}(d)\Big)$$

Output passed forward:

- freeze-adjusted ambient profile.

18.3 Heat-wave allocation and anomaly override (Eqs. 235–238)

Inputs:

- expected heat-wave starts S_m,
- expected heat-wave days H_m,
- mean heat-wave start day $\mu_{h,m}$,
- heat-wave mean anomaly $\Delta T_{h,m}$,
- heat-wave anomaly bounds $\Delta T^{\text{low}}_{h,m}, \Delta T^{\text{high}}_{h,m}$,
- heat-wave anomaly spread $\sigma_{\Delta T,h,m}$,
- freeze-adjusted ambient from 18.2.

Stepwise derivation:

$$\text{Step 1 (integer allocation):} \quad \{S_m, H_m\} \to \{n_{s,m}, n_{h,m}\} = \{\text{LR}(S_m), \text{LR}(H_m)\}.$$

$$\text{Step 2 (spell lengths):} \quad \text{cap } n_{s,m} \text{ for feasibility: } n_{s,m} \leftarrow \min\left(n_{s,m}, \left\lfloor \frac{L_m + 1}{4} \right\rfloor\right),$$

$$n^{\min}_{h,m} = 3n_{s,m}, \quad n^{\max}_{h,m} = L_m - (n_{s,m} - 1), \quad n_{h,m} \leftarrow \min\big(\max(n_{h,m}, n^{\min}_{h,m}),\, n^{\max}_{h,m}\big),$$

$$\ell_{h,m,q} = 3,\ q = 1, \ldots, n_{s,m}, \qquad e_m = n_{h,m} - 3n_{s,m},$$

$$\text{cyclic distribution (code-equivalent): for } n_{s,m} > 0,\ u_m = \left\lfloor \frac{e_m}{n_{s,m}} \right\rfloor,\ v_m = e_m - u_m n_{s,m},$$

$$\ell_{h,m,q} = \begin{cases} 3 + u_m + 1, & q \le v_m, \\ 3 + u_m, & q > v_m, \end{cases} \qquad q = 1, \ldots, n_{s,m},$$

$$\text{thus obtain } \boldsymbol{\ell}_{h,m} = (\ell_{h,m,1}, \ldots, \ell_{h,m,n_{s,m}}) \text{ with } \sum_q \ell_{h,m,q} = n_{h,m}.$$

$$\text{Step 3 (start-day base score):} \quad w_{h,m}(r) = \exp\left[-\frac{(r - \mu_{h,m})^2}{2s^2_{h,m}}\right], \qquad r = 1, \ldots, L_m.$$

Step 4 (recursive selected-start set): $\mathcal{S}_{h,m}^{(0)} = \varnothing$,

$$\mathcal{I}_{h,m}^{(k)} = \{1, \dots, L_m\} \setminus \mathcal{S}_{h,m}^{(k-1)},$$

$$\mathrm{score}_{h,m}^{(k)}(r) = w_{h,m}(r) \prod_{p \in \mathcal{S}_{h,m}^{(k-1)}} \left[1 - 0.65 \exp\left(-\frac{(r-p)^2}{2r_{h,m}^2}\right)\right], \qquad r \in \mathcal{I}_{h,m}^{(k)},$$

$$d_{h,m,k} \in \mathcal{I}_{h,m}^{(k)}, \qquad \mathrm{score}_{h,m}^{(k)}(d_{h,m,k}) = \max_{r \in \mathcal{I}_{h,m}^{(k)}} \mathrm{score}_{h,m}^{(k)}(r),$$

$$\mathcal{S}_{h,m}^{(k)} = \mathcal{S}_{h,m}^{(k-1)} \cup \{d_{h,m,k}\}, \qquad k = 1, \dots, n_{s,m}, \qquad \mathcal{S}_{h,m} = \mathcal{S}_{h,m}^{(n_{s,m})}.$$

Here $s_{h,m}$ is month-m start spread and $r_{h,m}$ is the repulsion length scale. At iteration k, "already selected" means exactly $\mathcal{S}_{h,m}^{(k-1)}$.

Step 5 (feasible contiguous spells): project raw starts to feasible $s_{h,m,q}$ so spells do not overlap and remain in m

Define raw starts by sorting the selected-start set: $d_{h,m,1}^{\mathrm{raw}} < \cdots < d_{h,m,n_{s,m}}^{\mathrm{raw}}$,

$$\{d_{h,m,1}^{\mathrm{raw}}, \dots, d_{h,m,n_{s,m}}^{\mathrm{raw}}\} = \mathcal{S}_{h,m}.$$

Thus $d_{h,m,q}^{\mathrm{raw}}$ is the q-th ordered raw start day (before feasibility clipping).

$$e_{h,m,q} = \begin{cases} 1, & q = 1, \\ s_{h,m,q-1} + \ell_{h,m,q-1} + 1, & q \geq 2, \end{cases} \qquad u_{h,m,q} = L_m - \left(\sum_{p=q}^{n_{s,m}} \ell_{h,m,p} + (n_{s,m} - q)\right) + 1,$$

$$s_{h,m,q} = \begin{cases} e_{h,m,q}, & d_{h,m,q}^{\mathrm{raw}} < e_{h,m,q}, \\ d_{h,m,q}^{\mathrm{raw}}, & e_{h,m,q} \leq d_{h,m,q}^{\mathrm{raw}} \leq u_{h,m,q}, \\ u_{h,m,q}, & d_{h,m,q}^{\mathrm{raw}} > u_{h,m,q}. \end{cases}$$

$$\mathcal{B}_{h,m,q} = \{\, r : \; s_{h,m,q} \leq r \leq s_{h,m,q} + \ell_{h,m,q} - 1 \,\}$$

Meaning: $\mathcal{B}_{h,m,q}$ is the set of all days covered by spell q (not a start-day set).

$$|\mathcal{B}_{h,m,q}| = \ell_{h,m,q}.$$

$$\mathcal{B}_{h,m}^{\mathrm{all}} = \bigcup_{q=1}^{n_{s,m}} \mathcal{B}_{h,m,q}, \qquad \left|\mathcal{B}_{h,m}^{\mathrm{all}}\right| = \sum_{q=1}^{n_{s,m}} \ell_{h,m,q} = n_{h,m}.$$

$$\text{and } q(d) = q \text{ when } r_m(d) \in \mathcal{B}_{h,m,q}.$$

Step 6 (spell anomaly sampling and ambient override): $\Delta T_{h,m,q} \in [a_{h,m}, b_{h,m}]$, $\Pr(u \leq \Delta T_{h,m,q} \leq v) = \int_u^v f_{h,m}$

$$\text{for } a_{h,m} \leq u \leq v \leq b_{h,m}.$$

$$a_{h,m} = \Delta T_{h,m}^{\mathrm{low}}, \quad b_{h,m} = \Delta T_{h,m}^{\mathrm{high}}, \quad \mu_{h,m}^{\star} = \mathrm{clip}(\Delta T_{h,m}, a_{h,m}, b_{h,m}), \quad \sigma_{h,m}^{\star} = \max(\sigma_{\Delta T,h,m}, \varepsilon),$$

where $\sigma_{\Delta T,h,m}$ is the month-m spell-anomaly spread from `heat_wave.spellAnomalyStd_C`$[m]$, with fallback

$$\sigma_{\Delta T,h,m} = \begin{cases} (\Delta T_{h,m}^{\mathrm{high}} - \Delta T_{h,m}^{\mathrm{low}})/4, & \Delta T_{h,m}^{\mathrm{high}} > \Delta T_{h,m}^{\mathrm{low}}, \\ 0, & \Delta T_{h,m}^{\mathrm{high}} \leq \Delta T_{h,m}^{\mathrm{low}}. \end{cases}$$

when configured spread is missing or nonpositive. The implementation floor is

$$\varepsilon = 10^{-3}\,{}^{\circ}\mathrm{C}$$

for the stochastic sampling branch. where $\Delta T_{h,m}$, $\Delta T^{\text{low}}_{h,m}$, and $\Delta T^{\text{high}}_{h,m}$ are month-m spell-anomaly-above-mean statistics from

`heat_wave.meanAnomalyAboveMonthlyMean_C`$[m]$, `heat_wave.minSpellAnomalyAboveMonthlyMean_C`$[m]$, `he`

Historical NOAA derivation with spell anomalies $a^{(h)}_{m,q}$: if $\mathcal{B}_{h,m,q}$ is the day set of historical spell q (grouped by spell start month m),

$$a^{(h)}_{m,q} = \frac{1}{|\mathcal{B}_{h,m,q}|} \sum_{t \in \mathcal{B}_{h,m,q}} \max\Big(0,\ T^{\text{avg}}_t - \bar{T}_{m(t)}\Big)\,,$$

where T^{avg}_t is NOAA daily mean proxy (TAVG if present, else $(TMAX + TMIN)/2$) and $m(t)$ is day t's calendar month index. So $\bar{T}_{m(t)}$ is the same $\bar{T}_m$ notation evaluated at day t's month.

Then

$$\Delta T_{h,m} = \frac{1}{N_m} \sum_{q=1}^{N_m} a^{(h)}_{m,q}, \qquad \Delta T^{\text{low}}_{h,m} = \min_q a^{(h)}_{m,q}, \qquad \Delta T^{\text{high}}_{h,m} = \max_q a^{(h)}_{m,q}.$$

$$g_{\text{std}}(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \qquad G_{\text{std}}(z) = \int_{-\infty}^{z} g_{\text{std}}(t)\,dt.$$

$$f_{h,m}(x) = \frac{g_{\text{std}}\left(\frac{x-\mu^\star_{h,m}}{\sigma^\star_{h,m}}\right)}{\sigma^\star_{h,m}\left[G_{\text{std}}\left(\frac{b_{h,m}-\mu^\star_{h,m}}{\sigma^\star_{h,m}}\right) - G_{\text{std}}\left(\frac{a_{h,m}-\mu^\star_{h,m}}{\sigma^\star_{h,m}}\right)\right]}, \quad a_{h,m} \le x \le b_{h,m},$$

$$\text{and } f_{h,m}(x) = 0 \text{ outside } [a_{h,m}, b_{h,m}].$$

Here z is a dimensionless standard-normal coordinate (z-score), and $Z_{h,m,q,t}$ is one random draw from that $N(0,1)$ distribution. So z itself is unbounded:

$$z \in (-\infty, \infty).$$

Bounds are enforced only through the accepted anomaly interval

$$a_{h,m} \le Y_{h,m,q,t} \le b_{h,m},$$

which is equivalent to the accepted z-interval

$$\frac{a_{h,m} - \mu^\star_{h,m}}{\sigma^\star_{h,m}} \le z \le \frac{b_{h,m} - \mu^\star_{h,m}}{\sigma^\star_{h,m}}.$$

Implementation-equivalent draw equations for spell q:

$$Z_{h,m,q,t} \sim g_{\text{std}}, \qquad Y_{h,m,q,t} = \mu^\star_{h,m} + \sigma^\star_{h,m} Z_{h,m,q,t}, \qquad t = 1, \ldots, 64.$$

$$\tau_{h,m,q} = \min\left\{t \in \{1, \ldots, 64\} :\ a_{h,m} \le Y_{h,m,q,t} \le b_{h,m}\right\}.$$

If $\tau_{h,m,q}$ exists:

$$\Delta T_{h,m,q} = Y_{h,m,q,\tau_{h,m,q}}.$$

If no accepted draw exists in 64 attempts:

$$\Delta T_{h,m,q} = \min\Big(\max\Big(\mu^\star_{h,m}, a_{h,m}\Big)\,,\ b_{h,m}\Big)\,.$$

$$\underbrace{f_{h,m}(x)}_{\text{distribution}} \Rightarrow \underbrace{\Delta T_{h,m,q}}_{\text{sample for spell } q} \Rightarrow \underbrace{T^{\text{th}}_{h,m,q} = \bar{T}_m + \Delta T_{h,m,q}}_{\text{spell threshold}} \Rightarrow \underbrace{T_\infty(d)}_{\text{daily ambient passed to 18.4}} .$$

$$\{T_{\infty,\text{base}}(d), \bar{T}_m, \Delta T_{h,m,q(d)}, T_{\infty,\text{preH}}(d)\} \to T_\infty(d)$$

$$= \begin{cases} \max\Big(T_{\infty,\text{base}}(d),\, \bar{T}_m + \Delta T_{h,m,q(d)},\, T_{\infty,\text{preH}}(d)\Big), & r_m(d) \in \mathcal{B}^{\text{all}}_{h,m}, \\ T_{\infty,\text{preH}}(d), & r_m(d) \notin \mathcal{B}^{\text{all}}_{h,m}. \end{cases}$$

Output passed forward:

- final daily ambient $T_\infty(d)$ used by all day-level thermal solves.

18.4 Daily heating and cooling loads (Eqs. 239–240)

Inputs:

- $T_\infty(d)$ from 18.3,
- heating and cooling setpoints $T_{h,\text{set}}, T_{c,\text{set}}$,
- day-specific bin thermal model terms (effective UA_{bin}).

Sequential feed:

$$\{T_\infty(d), T_{h,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},h}(d)$$
$$\{T_\infty(d), T_{c,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},c}(d)$$

Outputs passed forward:

- required day heating/cooling loads.

18.5 Capacity re-solve and duty application (Eqs. 241–246)

Inputs:

- $Q_{\text{req},h}(d)$, $Q_{\text{req},c}(d)$ from 18.4,
- final ambient $T_\infty(d)$ from 18.3,
- selected heating and cooling reference points.

Sequential feed:

$$\{T_\infty(d), \text{heating reference point}\} \to \{Q_{h,\text{cap}}(d), P_{h,\text{elec}}(d)\}$$

$$\{T_\infty(d), \text{cooling reference point}\} \to \{Q_{c,\text{cap}}(d), P_{c,\text{elec}}(d)\}$$

Here $Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d))$, $Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d))$, $P_{h,\text{elec}}(d) = P_h(T_\infty(d))$, and $P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d)$.

Explicit re-solve equations used here:

$$R_{\text{tube},g}(d) = \rho_{20}\Big[1 + \alpha\Big(\bar{T}_{w,g}(d) - 20\Big)\Big]\,\frac{L_{w,g}}{A_{w,g}}, \quad I_{\text{tube},g}(d) = \frac{V}{N_{s,g} R_{\text{tube},g}(d)},$$

$$P_{\text{tube},g}(d) = I^2_{\text{tube},g}(d) R_{\text{tube},g}(d), \quad P_{h,\text{coil}}(d) = \sum_g n_{\text{tube},g} P_{\text{tube},g}(d).$$

$$\Delta p_{\text{avail},x}(\dot{V}) = \max\left(\Delta p_{\text{shutoff},x}\left(1 - \frac{\dot{V}}{\dot{V}_{\text{free},x}}\right), 0\right), \quad R_x(\dot{V}) = \Delta p_{\text{sys},x}(\dot{V}) - \Delta p_{\text{avail},x}(\dot{V}),$$

with explicit system-pressure decomposition from the solver:

$$\Delta p_{\text{sys,ab}}(\dot{V}) = \Delta p_{\text{dist},c}(\dot{V}) + p_{\text{in},c}(\dot{V}),$$

$$\Delta p_{\text{dist},c} = \Delta p_{\text{header}} + \Delta p_{\text{header}\to\text{splitter}} + \Delta p_{\text{splitter body}} + \Delta p_{\text{connector fric}} + \Delta p_{\text{connector}\to\text{branch}},$$

$$R_{\text{inlet}}(p_{\text{in},c}) = \dot{m}_{\text{rem},N_{\text{seg}}+1}(p_{\text{in},c}) = 0,$$

$$\dot{m}_{\text{rel},s} = \min\left(C_d A_{h,s}\sqrt{2\rho_s \max(p_s - p_{\text{bed},s}, 0)}, \dot{m}_{\text{rem},s}\right),$$

$$p_{\text{bed},s} = p_{\text{out,bnd}} + \Delta p_{\text{Ergun},s},$$

$$\Delta p_{\text{Ergun},s} = L_s\left[\frac{150(1-\varepsilon)^2}{\varepsilon^3}\frac{\mu_s u_s}{d_p^2} + \frac{1.75(1-\varepsilon)}{\varepsilon^3}\frac{\rho_s u_s^2}{d_p}\right], \qquad u_s = \frac{\dot{m}_{\text{rel},s}}{\rho_s A_{\text{trib},s}}.$$

$$\Delta p_{\text{sys,hb}}(\dot{V}) = \Delta p_{\text{dist,common}}(\dot{V}) + \Delta p_{\text{path,ctrl}}(\dot{V}),$$

$$\Delta p_{\text{path},g} = \Delta p_{\text{dist,private},g} + \Delta p_{\text{heater,tube},g} + \Delta p_{\text{heater,coil},g} + \Delta p_{\text{heater}\to\text{aer},g} + p_{\text{in,aer},g},$$

$$\Delta p_{\text{path,ctrl}} = \Delta p_{\text{path},g_{\text{ctrl}}}, \qquad \Delta p_{\text{path},g_{\text{ctrl}}} \geq \Delta p_{\text{path},g} \ \text{ for every } g.$$

with

$$\dot{V}_{\text{free},x} = \begin{cases} \dot{V}_{\text{rated},x}\dfrac{\Delta p_{\text{shutoff},x}}{\Delta p_{\text{shutoff},x} - \Delta p_{\text{rated},x}}, & \Delta p_{\text{shutoff},x} > \Delta p_{\text{rated},x}, \\ \dot{V}_{\text{rated},x}, & \text{otherwise}, \end{cases}$$

This follows from a two-point linear fan curve through

$$(\dot{V}, \Delta p) = (0, \Delta p_{\text{shutoff},x}), \qquad (\dot{V}, \Delta p) = (\dot{V}_{\text{rated},x}, \Delta p_{\text{rated},x}),$$

with slope

$$m_x = \frac{\Delta p_{\text{rated},x} - \Delta p_{\text{shutoff},x}}{\dot{V}_{\text{rated},x}}, \qquad \Delta p_x(\dot{V}) = \Delta p_{\text{shutoff},x} + m_x\dot{V}.$$

Setting $\Delta p_x(\dot{V}_{\text{free},x}) = 0$ yields the expression above. If $\Delta p_{\text{shutoff},x} \leq \Delta p_{\text{rated},x}$, that extrapolation is not physically reliable, so code falls back to $\dot{V}_{\text{free},x} = \dot{V}_{\text{rated},x}$. In plain terms: $\Delta p_{\text{shutoff},x}$ and $\Delta p_{\text{rated},x}$ are two pressure-rise points (not absolute pressures) on the same blower curve, at $\dot{V} = 0$ and $\dot{V} = \dot{V}_{\text{rated},x}$. Those two points define the linear slope used to compute $\Delta p_{\text{avail},x}(\dot{V})$ and $\dot{V}_{\text{free},x}$. Code construction sequence: if $\dot{V}_{\text{free},x}$ is provided in configuration, the active line is directly $(0, \Delta p_{\text{shutoff},x}) \to (\dot{V}_{\text{free},x}, 0)$; otherwise $\dot{V}_{\text{free},x}$ is first inferred from $(\dot{V}_{\text{rated},x}, \Delta p_{\text{rated},x})$ and $\Delta p_{\text{shutoff},x}$, and then the same shutoff-to-free line is used for $\Delta p_{\text{avail},x}$. when explicit free-air flow is not provided.

$$\mathcal{F}_x(d) = \left\{\dot{V} \in [0, \dot{V}_{\text{req},x}(d)] \ : \ R_x(\dot{V}) \leq 0\right\},$$

$$\dot{V}_{\text{feas},x}(d) = \begin{cases} \max \mathcal{F}_x(d), & \mathcal{F}_x(d) \neq \varnothing, \\ 0, & \mathcal{F}_x(d) = \varnothing, \end{cases}$$

$$\dot{V}_{\text{ach},x}(d) = \begin{cases} \dot{V}_{\text{req},x}(d), & \text{if blower } x \text{ is disabled, fan data are unavailable, or } R_x(\dot{V}_{\text{req},x}(d)) \leq 0, \\ \dot{V}_{\text{feas},x}(d), & \text{otherwise.} \end{cases}$$

$$\Delta p_{\mathrm{op},x}(d) = \begin{cases} 0, & \Delta p_{\mathrm{sys},x}(\dot{V}_{\mathrm{ach},x}(d)) \le 0, \\ \Delta p_{\mathrm{sys},x}(\dot{V}_{\mathrm{ach},x}(d)), & 0 < \Delta p_{\mathrm{sys},x}(\dot{V}_{\mathrm{ach},x}(d)) < \Delta p_{\mathrm{avail},x}(\dot{V}_{\mathrm{ach},x}(d)), \\ \Delta p_{\mathrm{avail},x}(\dot{V}_{\mathrm{ach},x}(d)), & \Delta p_{\mathrm{sys},x}(\dot{V}_{\mathrm{ach},x}(d)) \ge \Delta p_{\mathrm{avail},x}(\dot{V}_{\mathrm{ach},x}(d)). \end{cases}$$

$$P_{\mathrm{shaft},x}(d) = \dot{V}_{\mathrm{ach},x}(d)\Delta p_{\mathrm{op},x}(d),$$

$$P_{x,\mathrm{elec}}(d) = \min\left(P_{\mathrm{rated},x},\, P_{\mathrm{idle},x} + \frac{P_{\mathrm{shaft},x}(d)}{\eta_x}\right),$$

and when η_x is unavailable in configuration, code falls back to

$$P_{x,\mathrm{elec}}(d) = \min\left(P_{\mathrm{rated},x}, P_{\mathrm{idle},x} + (P_{\mathrm{rated},x} - P_{\mathrm{idle},x}) \min\left(\max\left(\frac{\dot{V}_{\mathrm{ach},x}(d)}{\dot{V}_{\mathrm{free},x}}, 0\right), 1\right)\right).$$

Equivalent implemented piecewise logic:

$$P_{x,\mathrm{elec}}(d) = \begin{cases} 0, & \text{if blower } x \text{ is disabled,} \\ & \text{or } \dot{V}_{\mathrm{ach},x}(d) \le 0 \\ \min\left(P_{\mathrm{rated},x},\, P_{\mathrm{idle},x} + \frac{P_{\mathrm{shaft},x}(d)}{\eta_x}\right), & \text{if } \eta_x \text{ is provided,} \\ \min\left(P_{\mathrm{rated},x}, P_{\mathrm{idle},x} + (P_{\mathrm{rated},x} - P_{\mathrm{idle},x}) \min\left(\max\left(\frac{\dot{V}_{\mathrm{ach},x}(d)}{\dot{V}_{\mathrm{free},x}}, 0\right), 1\right)\right), & \text{otherwise.} \end{cases}$$

Interpretation: $P_{\mathrm{idle},x}$ is baseline blower electrical overhead at low flow (motor/controller/parasitics), not a constant always-on penalty. In code it is read from `idlePower_W`, clamped nonnegative, and applied only when blower is enabled and solved flow is positive.

$$Q_{\mathrm{to\,bed},h}(T_\infty(d)) = \sum_{b=1}^{N_b} \sum_{s=1}^{N_{\mathrm{seg}}} \left(Q_{\mathrm{wall},b,s} + Q_{\mathrm{jet},b,s} - Q_{\mathrm{evap},b,s}\right).$$

$$\dot{V}_{\mathrm{rated}} = 0.00047194745\, V_{\mathrm{rated,CFM}}, \quad \dot{m}_{\mathrm{rated}}(d) = \rho_{\mathrm{air}}(T_\infty(d), p_\infty)\, \dot{V}_{\mathrm{rated}},$$

$$\dot{Q}_{\mathrm{rated,sens}}(d) = \max\left(\dot{Q}_{\mathrm{rated,total}} - \dot{Q}_{\mathrm{rated,lat}}(d),\, 0\right),$$

$$T_{\mathrm{target,derived}}(d) = T_\infty(d) - \frac{\dot{Q}_{\mathrm{rated,sens}}(d)}{\dot{m}_{\mathrm{rated}}(d) c_p(d)} \quad \text{when } \dot{m}_{\mathrm{rated}}(d) c_p(d) > 0,$$

$$T_{\mathrm{target}}(d) = \begin{cases} T_{\mathrm{target,derived}}(d), & \text{if } \chi_{\mathrm{derive}} = 1 \text{ and } T_{\mathrm{target,derived}}(d) \text{ is finite,} \\ T_{\mathrm{target,cfg}}, & \text{if a configured target is provided,} \\ T_\infty(d), & \text{otherwise,} \end{cases}$$

$$\dot{m}(d) = \rho_{\mathrm{air}}(T_\infty(d), p_\infty)\, \dot{V}_{\mathrm{ach,ab}}(d), \qquad \dot{m}_{\mathrm{dry}}(d) = \frac{\dot{m}(d)}{1 + w_{\mathrm{in}}(d)}.$$

with $\dot{V}_{\mathrm{ach,ab}}(d)$ obtained from the assist-blower residual closure $R_{\mathrm{ab}}(\dot{V}) = \Delta p_{\mathrm{sys,ab}}(\dot{V}) - \Delta p_{\mathrm{avail,ab}}(\dot{V})$ over $[0, \dot{V}_{\mathrm{req,ab}}(d)]$.

$$\dot{Q}_{\mathrm{spot,req}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\mathrm{target}}(d), 0), \quad \dot{Q}_{\mathrm{spot,act}}(d) = \min\left(\dot{Q}_{\mathrm{spot,req}}(d), \dot{Q}_{\mathrm{spot,max}}(d)\right),$$

$$\dot{Q}_{\text{spot,max}}(d) = \begin{cases} Q_{\text{spot,max,cfg}}, & \text{if } \texttt{maxCoolingCapacity_W} \text{ is provided,} \\ \max\Big(\dot{Q}_{\text{rated,total}} - \dot{Q}_{\text{rated,lat}}(d),\, 0\Big), & \text{if rated-spec path is used,} \\ +\infty, & \text{otherwise,} \end{cases}$$

$$\dot{Q}_{\text{rated,total}} = 0.29307107\, C_{\text{rated,BTU/h}},$$
$$\dot{m}_{\text{cond,rated}} = \frac{0.473176473\, D_{\text{rated,pint/day}}}{86400},$$
$$\dot{Q}_{\text{rated,lat}}(d) = \dot{m}_{\text{cond,rated}}\, h_{fg}(T_\infty(d)).$$

$$\dot{Q}_{\text{sensible}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\text{supply,actual}}(d), 0) = \dot{Q}_{\text{spot,act}}(d).$$

$$P_{\text{in}}(d) \equiv P_{\text{spot cooler}}(d).$$

$$\text{COP} = \frac{\dot{Q}_{\text{cool}}}{P_{\text{in}}(d)},$$
$$\text{COP}_{\text{sensible}} = \frac{\dot{Q}_{\text{sensible}}(d)}{P_{\text{in}}(d)},$$
$$\text{COP} = \frac{\text{EER}}{3.412}.$$

$$P_{\text{spot cooler}}(d) = \frac{\dot{Q}_{\text{spot,act}}(d)}{\text{COP}}, \quad P_{\text{assist blower}}(d) = P_{\text{ab,elec}}(d), \quad P_h(T_\infty(d)) = P_{h,\text{coil}}(d) + P_{\text{hb,elec}}(d).$$

In this model, COP is used as sensible COP because $\dot{Q}_{\text{spot,act}}(d)$ is sensible load.

$$Q_{\text{to bed},c}(T_\infty(d)) = \sum_{s=1}^{N_{\text{seg}}} \left(Q_{\text{wall},s} + Q_{\text{jet},s} - Q_{\text{evap},s}\right).$$

So "re-solve for $Q_{\text{to bed}}$" means: re-run these day-ambient equations, solve the scalar closure residuals (blower flow and pressure closures), then evaluate the segment energy sums. It is *not* a matrix-residual iteration.

Variable definitions for the equations above:

- Indices: d day index; g heater-tube group; b aeration branch; s axial segment.
- Integer counts: $n_{\text{tube},g}$ tubes in group g; N_b total branches; N_{seg} total segments per branch.
- Electrical/wire terms: ρ_{20} resistivity at 20 °C; α wire temperature coefficient; $\bar{T}_{w,g}(d)$ representative wire temperature; $L_{w,g}$ wire length; $A_{w,g}$ wire cross-sectional area; V supply voltage; $N_{s,g}$ series count; $R_{\text{tube},g}(d)$ tube resistance; $I_{\text{tube},g}(d)$ tube/string current; $P_{\text{tube},g}(d)$ tube/string power; $P_{h,\text{coil}}(d)$ total heater-coil power; $P_h(T_\infty(d))$ total heating power after adding heating-blower electrical power.
- Heating bed-transfer terms: $\dot{m}_{\text{mean},b,s}$ mean segment flow; $c_{p,b,s}$ segment specific heat; $T_{\text{air,in},b,s}, T_{\text{air,out},b,s}$ segment inlet/outlet air temperatures; $\dot{m}_{\text{rel},b,s}$ released jet flow; $T_{\text{rel},b,s}$ jet/release temperature; $T_{\text{bed,ref}}$ bed reference temperature; $\dot{m}_{\text{evap},b,s}$ evaporation mass flow; $h_{fg,b,s}$ latent heat; $Q_{\text{wall},b,s}$, $Q_{\text{jet},b,s}$, $Q_{\text{evap},b,s}$ segment wall/jet/evaporation terms; $Q_{\text{to bed},h}(T_\infty(d))$ net signed heating-mode bed transfer.

- Spot-cooling terms: $\dot{V}_{\text{ach,ab}}(d)$ achieved assist-blower flow from the blower/network residual solve; $\dot{m}(d) = \rho_{\text{air}}(T_\infty(d), p_\infty)\dot{V}_{\text{ach,ab}}(d)$ day moist-air mass flow; $w_{\text{in}}(d)$ inlet humidity ratio; $\dot{m}_{\text{dry}}(d) = \dot{m}(d)/(1 + w_{\text{in}}(d))$ day dry-air flow; $c_p(d)$ day effective specific heat; $T_\infty(d)$ day ambient; $T_{\text{target,cfg}}$ configured supply target (`targetSupplyAir_C`); χ_{derive} derivation toggle (`deriveTargetFromRatedSpecs`); $V_{\text{rated,CFM}}$ rated airflow; $\dot{V}_{\text{rated}}$ rated airflow in SI; p_∞ ambient pressure; ρ_{air} density function; $\dot{m}_{\text{rated}}(d)$ rated mass flow; $T_{\text{target,derived}}(d)$ target derived from rated sensible capacity; $T_{\text{target}}(d)$ effective target used in load calculation; $\dot{Q}_{\text{spot,req}}(d)$ requested sensible load; $\dot{Q}_{\text{spot,max}}(d)$ max sensible capacity; $\dot{Q}_{\text{spot,act}}(d)$ capped actual sensible load; $T_{\text{supply,actual}}(d)$ actual post-cap supply-air temperature; $\dot{Q}_{\text{sensible}}(d) = \dot{m}(d)c_p(d)\max(T_\infty(d) - T_{\text{supply,actual}}(d), 0)$ sensible cooling rate (equal to $\dot{Q}_{\text{spot,act}}(d)$ here); $P_{\text{in}}(d) \equiv P_{\text{spot cooler}}(d)$ electrical input used in COP definitions; COP coefficient of performance; $P_{\text{spot cooler}}(d)$ cooler electric power.

- Blower/closure terms: $x \in \{\text{hb}, \text{ab}\}$ (heating blower, assist blower); $\Delta p_{\text{shutoff},x}$ is the datasheet shutoff pressure (maximum static pressure at zero flow; `shutoffPressure_inH2O` converted to Pa); $\dot{V}_{\text{rated},x}$ is the datasheet rated flow (`ratedFlow_CFM`, converted to SI); $\Delta p_{\text{rated},x}$ is the pressure paired with that same rated point (`ratedPressure_inH2O`, converted to Pa); $\dot{V}_{\text{free},x}$ is free-delivery flow (explicit or inferred from those rated/shutoff values); $\Delta p_{\text{avail},x}(\dot{V})$ available blower pressure at trial $\dot{V}$; $\Delta p_{\text{sys,ab}} = \Delta p_{\text{dist},c} + p_{\text{in},c}$; $\Delta p_{\text{sys,hb}} = \Delta p_{\text{dist,common}} + \Delta p_{\text{path,ctrl}}$; $\Delta p_{\text{dist},c}$ is cooling distribution loss sum; $p_{\text{in},c}$ is closed-end aeration inlet pressure; $p_{\text{in,aer},g}$ is group-g heating aeration inlet pressure; $p_{\text{out,bnd}}$ is bed outlet-boundary pressure; $\Delta p_{\text{Ergun},s}$ is segment porous-backpressure term; R_{inlet} is closed-end inlet residual; g_{ctrl} is the heating group with the largest private-path pressure drop; $R_x(\dot{V})$ is blower flow residual; $\dot{V}_{\text{req},x}(d)$ requested flow; $\mathcal{F}_x(d)$ feasible-flow set; $\dot{V}_{\text{feas},x}(d)$ maximum feasible flow helper; $\dot{V}_{\text{ach},x}(d)$ achieved flow; $\Delta p_{\text{op},x}(d)$ bounded operating pressure; $P_{\text{shaft},x}(d)$ shaft power; $P_{\text{idle},x}$ baseline low-flow electrical overhead from `idlePower_W` (nonnegative in code, applied only when blower is enabled and $\dot{V}_{\text{ach},x} > 0$); $P_{\text{rated},x}$ rated power cap; η_x total efficiency; $P_{x,\text{elec}}(d)$ blower electrical power; $P_{\text{assist blower}}(d) = P_{\text{ab,elec}}(d)$.

- Fan-curve interpretation: $\Delta p_{\text{shutoff},x}$ and $\Delta p_{\text{rated},x}$ are static pressure-rise values on the same blower curve (Pa), not absolute pressures. They are the $\dot{V} = 0$ and $\dot{V} = \dot{V}_{\text{rated},x}$ endpoints used to define the linear pressure-vs-flow relation.

- Cooling bed-transfer terms: $Q_{\text{wall},s}$, $Q_{\text{jet},s}$, $Q_{\text{evap},s}$ segment contributions; $Q_{\text{to bed},c}(T_\infty(d))$ net signed cooling-mode bed transfer (usually negative).

Derivation relevance (short):

- Electrical equations come from temperature-corrected resistivity, Ohm's law, and Joule heating.

- $Q_{\text{to bed}}$ equations are direct first-law sums over segment sensible and latent terms.

- Spot-cooler load uses $\dot{m}c_p\Delta T$, then caps by available capacity and converts to electric power via COP.

- Blower flow comes from residual matching $\Delta p_{\text{sys},x} = \Delta p_{\text{avail},x}$ solved by scalar bisection over $[0, \dot{V}_{\text{req},x}]$. Here $\Delta p_{\text{sys},x}$ includes distribution losses plus closed-end aeration inlet pressure, and the inlet term includes segmentwise Ergun porous-backpressure; blower electrical power then follows from $P_{\text{shaft},x} = \dot{V}_{\text{ach},x}\Delta p_{\text{op},x}$, efficiency conversion, and rated-power cap.

- Heating electrical power uses $P_h = P_{h,\text{coil}} + P_{\text{hb,elec}}$; cooling electrical power uses $P_{c,\text{elec}} = P_{\text{spot cooler}} + P_{\text{ab,elec}}$.

$$\{Q_{\text{req},h}(d), Q_{h,\text{cap}}(d)\} \to D_h^*(d) \to D_h(d) = \text{clip}_{[0,1]}(D_h^*)$$
$$\{Q_{\text{req},c}(d), Q_{c,\text{cap}}(d)\} \to D_c^*(d) \to D_c(d) = \text{clip}_{[0,1]}(D_c^*)$$
$$\{D_h(d), D_c(d), Q_{h,\text{cap}}(d), Q_{c,\text{cap}}(d)\} \to Q_{\text{bed}}(d)$$

Outputs passed forward:

- applied bed thermal load $Q_{\text{bed}}(d)$,
- applied duty fraction (D_h or D_c),
- mode-specific electric power ($P_{h,\text{elec}}$ or $P_{c,\text{elec}}$).

18.6 Daily temperatures, electricity, cost, and water (Eqs. 247–251)

Inputs:

- $Q_{\text{bed}}(d)$, duty, and mode power from 18.5,
- $T_\infty(d)$ from 18.3,
- two-node conductance parameters and f_b,
- electricity unit price c_{elec},
- nominal daily water-loss rate $\dot{m}_{\text{water},24h}$.

Sequential feed:

$$\{Q_{\text{bed}}(d), T_\infty(d), UA_{b,\infty}, UA_{t,\infty}, UA_{bt}, f_b\} \to \{T_b(d), T_t(d)\}$$
$$\{P_{h,\text{elec}}(d), D_h(d)\} \to E_h(d), \qquad \{P_{c,\text{elec}}(d), D_c(d)\} \to E_c(d) \to E_{\text{day}}$$
$$\{E_{\text{day}}, c_{\text{elec}}\} \to C(d)$$
$$\{\dot{m}_{\text{water},24h}, D(d)\} \to m_{\text{water}}(d)$$

Final annual aggregation feed:

$$\sum_{d=1}^{365} \{E_h(d), E_c(d), C(d), m_{\text{water}}(d)\} \to \{\text{annual kWh, annual USD, annual water}\}$$

8 Clear Report Narrative You Can Follow

Use this 8-step order when drafting your report section:

1. **State the objective**: "Section 18 converts monthly climate statistics into a representative daily operating year and computes day-level thermal duty, electricity, cost, and water."
2. **Define climate reference values**: monthly means $\bar{T}_m$ and spreads σ_m from NOAA-derived arrays.
3. **Describe baseline reconstruction**: explain periodic Gaussian kernel regression and why circular distance removes Dec/Jan discontinuity.

4. **Describe event overlays**: explain freeze-day and heat-wave day integer allocation (largest remainder), day placement (Gaussian-centered with repulsion), and spell-level bounded-normal anomaly sampling for heat-wave intensity.

5. **Move to daily thermal demand**: compute required heating/cooling from day ambient and setpoints.

6. **Explain ambient-specific capacity re-solving**: heating and cooling reference points are re-evaluated at each day ambient before duty is computed.

7. **Explain bounded operation**: duty is clipped to $[0, 1]$, applied bed load is signed by mode, and unmet load is tracked separately when needed.

8. **Close with economics and resources**: convert day power to kWh, then to cost (USD/day), and duty-scaled water replacement, then aggregate by month and year.

9 Draft Paragraph Template (Copy And Adapt)

Section 18 implements an equation-driven annual operating layer that maps monthly climate reference values to daily ambient conditions, then computes daily heating or cooling requirements and operating costs. First, monthly mean and standard-deviation reference values are converted into a smooth 365-day baseline using periodic Gaussian kernel regression (Eqs. 229–230). Next, freeze days and heat-wave spells are allocated by largest-remainder rounding and positioned with Gaussian timing weights plus repulsion to avoid unrealistic clustering; heat-wave spell magnitudes are then drawn from seeded bounded-normal monthly distributions (Eqs. 231–238). For each day, required heating/cooling loads are evaluated from the day ambient and setpoints (Eqs. 239–240). The model then re-solves heating and cooling reference operating points at that specific ambient, computes required duty, clips applied duty to physical bounds, and applies signed bed load by mode (Eqs. 241–246). Bottom and top bed-node temperatures are solved from the same two-node steady model used elsewhere (Eq. 247), and daily electricity, cost, and water replacement are computed and aggregated to monthly and annual totals (Eqs. 248–251).

10 Quality Checks To Mention In The Report

- Duty clipping enforces physical feasibility of operation (`dutyApplied` in $[0, 1]$).
- Unmet load is explicitly quantified (`unmetLoad_W`, `annualUnmet_kWh_equivalent`).
- Heating and cooling power are ambient-specific daily re-solves, not fixed constants.
- Freeze/heat-wave event totals are auditable at monthly and annual levels.
- Heat-wave spell anomalies are bounded by configured monthly low/high arrays and reproducible via fixed RNG seed.

11 Current Run Snapshot (Example Numbers)

From `Heat Transfer Study/outputs/year_round/study_summary.txt` in this repository state:

- Allocated annual freeze days: 120

- Allocated annual heat-wave starts: 4
- Allocated annual heat-wave days: 15
- Annual electricity: 660.9 kWh
- Annual operating cost: 190.01 USD
- Annual water replacement: 3.4 kg

Use these only as run-specific examples; your final report values should come from the run you cite.